HELWAN NATIONAL UNIVERSITY

Robotics and Mechatronics Department

Pneumatic Circuit Design and Simulation

Subject: Hydraulic & Pneumatic Control

Course Code: MEC2307

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Introduction

This report presents the design and simulation of pneumatic control systems for two different applications. Each application represents a practical system that uses pneumatic cylinders to perform a specific task.

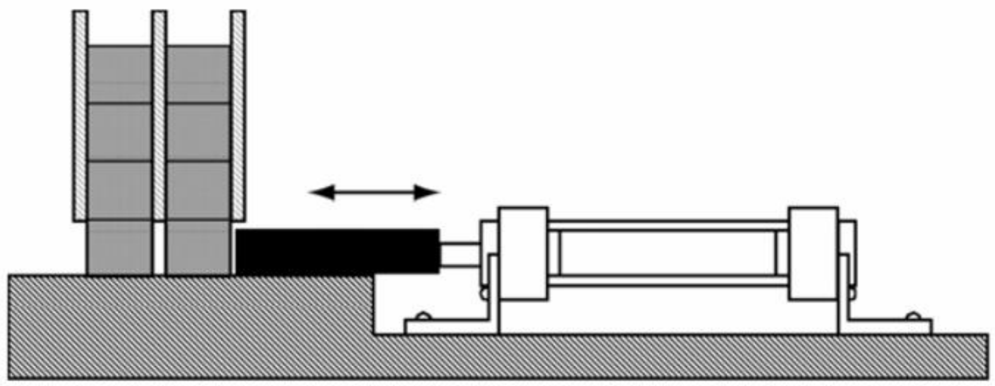
The required pneumatic circuits were designed using standard symbols and then implemented and tested using FluidSIM-P software to verify their operation. For each system, the circuit behavior is analyzed through simulation, and the corresponding control logic is developed where applicable.

The two applications covered in this report are:

Application 1 — Pneumatic Industrial Transport System: A single-acting cylinder pushes goods from a storage shelf onto a transfer belt, controlled by a 3/2 normally closed pushbutton valve with spring return.

Application 2 — Automatic Pneumatic Door Control: A double-acting cylinder opens and closes a vehicle door, controlled by a 5/2 solenoid-operated valve with relay-based control logic incorporating a proximity sensor, end-position sensor, and timer for automatic and safe operation.

Application 1: Pneumatic Industrial Transport System



Application Overview

This application involves the design and simulation of a pneumatic control system for a simplified industrial transport mechanism. A single-acting pneumatic cylinder is used to push goods from a storage shelf onto a transfer belt.

When the operator presses a pushbutton, compressed air is supplied to the cylinder, causing it to extend and push one item forward. When the pushbutton is released, the valve returns to its normal position, and the cylinder retracts automatically using its internal spring.

The system is designed using a 3/2 normally closed pushbutton valve with spring return and implemented using FluidSIM-P software to verify its operation.

Part (a) — Manual Pneumatic Circuit

Introduction

In this part, the pneumatic circuit is designed using a 3/2 pushbutton-actuated, spring-return directional control valve to control a single-acting cylinder.

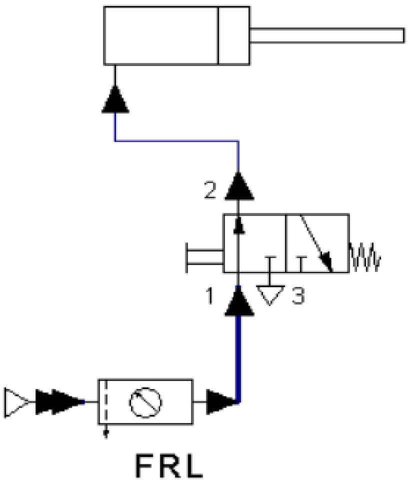
When the operator presses the pushbutton, compressed air flows into the cylinder, extending the piston rod and pushing the goods onto the conveyor. When the pushbutton is released, the valve returns to its normal position, exhausting the air and allowing the internal spring of the cylinder to retract the piston automatically.

Components Used

Component	Description
Single-acting cylinder	Extends under compressed air pressure, retracts using internal spring
3/2 pushbutton valve (normally closed, spring return)	Supplies air to the cylinder when pressed, exhausts air when released
FRL unit (Filter–Regulator–Lubricator)	Conditions compressed air before entering the system
Compressed air source	Provides pressurized air for the system

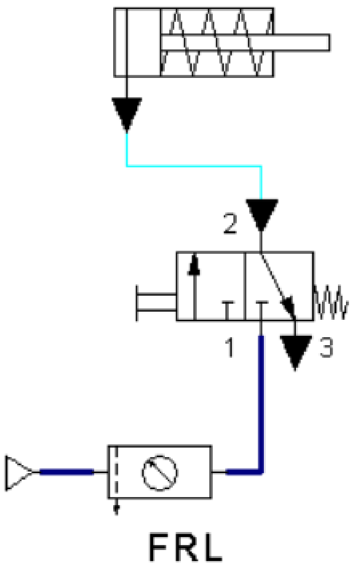
Circuit Operation

Extension Stroke — Goods Pushed



When the operator presses the pushbutton, the 3/2 directional control valve shifts from its normal position, connecting port 1 (pressure) to port 2 (output). Compressed air flows from the supply through the FRL unit and into the cylinder. This causes the piston to extend, pushing the goods from the shelf onto the transfer belt. The airflow path in the simulation confirms that pressurized air is delivered to the cylinder during this stroke.

Return Stroke — Goods Clear



When the pushbutton is released, the spring returns the valve to its normal position, connecting port 2 to port 3 (exhaust). The compressed air inside the cylinder is released to the atmosphere, and the internal spring of the cylinder pulls the piston rod back to its original position. This retracts the cylinder automatically, preparing the system for the next operation.

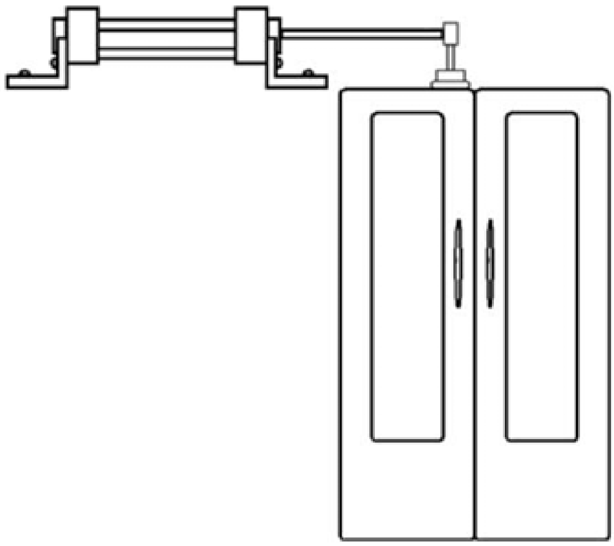
Conclusion — Application 1

In this application, a pneumatic control system was successfully designed and simulated for an industrial transport mechanism using a single-acting cylinder.

The use of a 3/2 normally closed pushbutton valve provides a simple and effective control method — pressing the button extends the cylinder to push the goods, and releasing it allows the cylinder to retract automatically via its internal spring.

The circuit proved to be efficient, reliable, and suitable for repetitive industrial operations where automatic return is required without additional control elements. The simulation in FluidSIM-P verified correct operation of both the extension and return strokes, confirming that the system performs according to the design requirements.

Application 2: Automatic Pneumatic Door Control



Application Overview

This application presents the design and simulation of a pneumatic control system used to automatically operate a door using a double-acting cylinder. The system is designed to enhance safety and automation by incorporating a proximity sensor, end-position sensor, timer, and relay-based control logic.

The system operates based on detecting the presence of a person near the door. When a person is detected, the door opens automatically and remains open as long as the person is present. Once the person leaves, the system waits for a specified delay of 5 seconds before automatically closing the door. This behavior ensures both user safety and energy efficiency, preventing unnecessary opening and closing actions.

Introduction

In this application, a double-acting pneumatic cylinder is used to open and close a door. The system is controlled using a 5/2 directional control valve operated by two solenoids (X and Y). Unlike manual systems, this circuit uses several sensing and timing elements to achieve fully automatic operation.

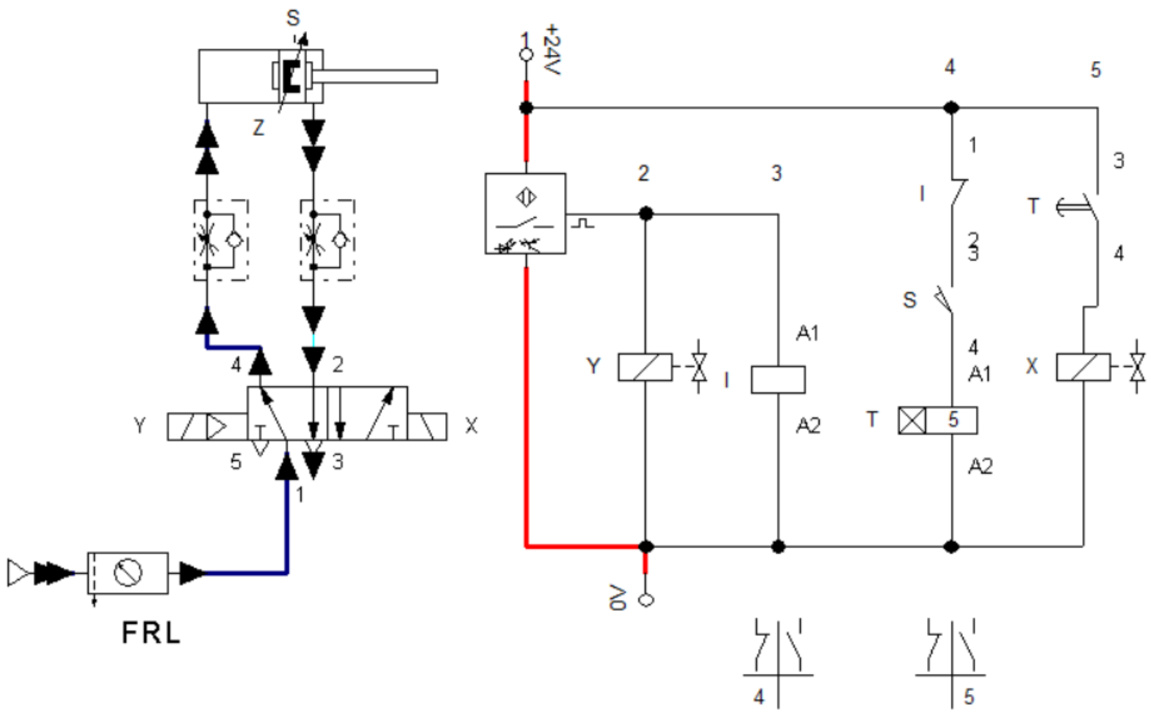
Components Used

Component	Description
Double-acting cylinder	Provides motion for opening and closing the door
5/2 directional control valve	Controls airflow direction to extend and retract the cylinder
Solenoid Y	Energizes valve to extend cylinder — door opens
Solenoid X	Energizes valve to retract cylinder — door closes
Optical proximity sensor	Detects presence of a person near the door

End-position sensor (S)	Detects when the cylinder is fully extended
Timer (T)	Provides 5-second delay before closing
Relay Y	Prevents timer activation while sensor is active
Relay X	Activates return stroke after timer finishes
FRL unit	Filters, regulates, and lubricates compressed air
Power supply (24V DC)	Supplies the electrical control circuit

System Operation

1. Extension Stroke — Door Opens



When a person approaches the door, the optical proximity sensor is activated, sending a signal to the control circuit. This signal energizes relay Y, which activates solenoid Y. The 5/2 valve shifts position, compressed air flows into the cylinder, the piston extends, and the door opens. Relay Y simultaneously disables the timer, ensuring the door does not close while someone is present. As long as the sensor continues detecting motion, relay Y remains energized and the door stays fully open.

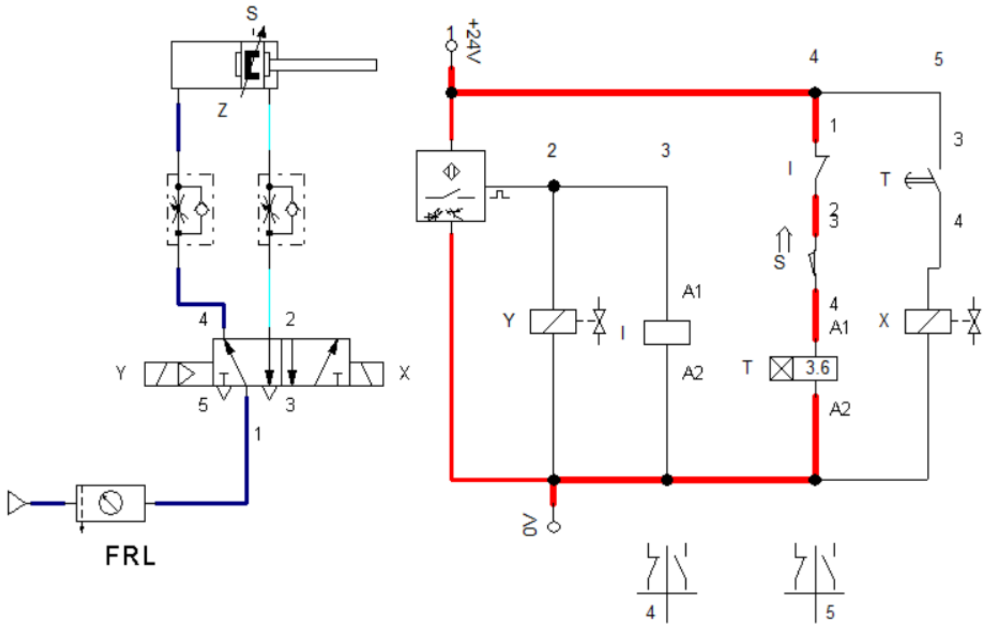
2. Holding Condition — Door Remains Open

While the proximity sensor is still detecting a person, relay Y stays energized, the timer remains off, solenoid X is not activated, and the cylinder stays extended. This creates a safe holding condition, ensuring the door does not close on a person.

3. Delay Activation — Sensor Stops Detecting

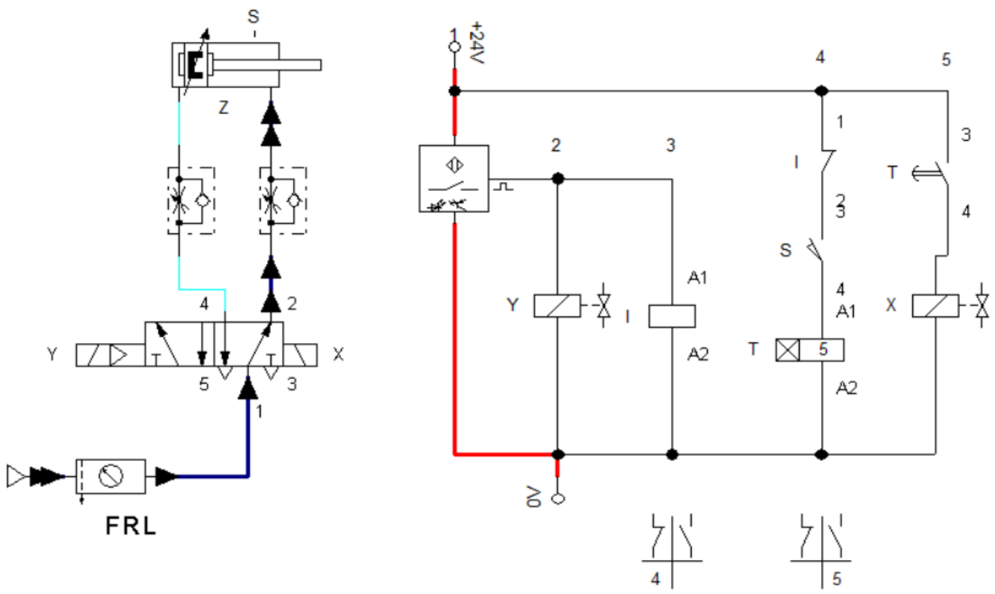
When the person leaves the detection area, the proximity sensor turns off and relay Y is de-energized, activating the timer circuit. However, the door does not close immediately. The end-position sensor (S) first confirms that the cylinder is fully extended before triggering the timer (T).

4. Timer Operation (5-Second Delay)



The timer begins counting for 5 seconds. During this period the cylinder remains extended and the door stays open, providing safe exit time and improving system usability.

5. Return Stroke — Door Closes



After the timer reaches 5 seconds, the timer contact closes and relay X is energized, activating solenoid X. The 5/2 valve switches direction, airflow reverses, the cylinder retracts, and the door closes automatically.

6. System Reset

After the cylinder fully retracts, the end-position sensor resets, the timer resets, and the system returns to its initial state — ready for the next detection cycle.

Control Logic Explanation

Relay Y — Priority Control

Relay Y is activated directly by the proximity sensor. It keeps the door open and blocks timer operation, ensuring the door never closes while someone is present.

Timer T — Delay Control

The timer is activated only when the sensor is off and the cylinder is confirmed fully extended. This ensures the door closes only after a safe delay has elapsed.

Relay X — Return Control

Relay X is activated after the timer finishes, controlling the closing motion of the cylinder. This ensures a controlled and time-delayed door closing sequence.

Conclusion — Application 2

In this application, a fully automated pneumatic door control system was successfully designed and simulated using FluidSIM.

The system demonstrates the integration of pneumatics (cylinder and valve), sensors (proximity and position), and electrical control (relays and timer). Key advantages of this design include automatic operation without human input, safety enhancement ensuring the door does not close while someone is present, time-delay closing for user convenience, and reliable control using relay logic.

The simulation confirmed correct operation of all stages: door opening when motion is detected, door holding while presence exists, timed delay after detection ends, and automatic door closing. This system represents a practical implementation of industrial automation using electro-pneumatic control.

Overall Conclusion

This report presented the design and simulation of two pneumatic control systems using FluidSIM-P software. Each application demonstrated a different aspect of pneumatic circuit design, from basic single-acting cylinder control to a fully automated electro-pneumatic system.

Application 1 demonstrated how a single-acting cylinder can effectively push goods using a simple 3/2 pushbutton valve with spring return. The simplicity of the design makes it ideal for repetitive industrial transport operations where automatic return is required without additional control elements.

Application 2 showed the advantage of combining a double-acting cylinder with sensor-based relay logic to achieve safe and fully automatic door control. The proximity sensor, end-position sensor, timer, and relay interlocking together ensure the door responds intelligently to the presence of people — opening automatically, holding safely, and closing only after a guaranteed delay.

Overall, the simulations confirmed that both circuits operate correctly according to their design specifications. The use of FluidSIM-P provided a reliable platform for verifying circuit behavior before physical implementation.